Customer Stories / Aerospace & Satellite

2024

Scaling HPC in 6 Months for Aerospace Modeling Using AWS with Hypersonix Launch Systems

Learn how aerospace company Hypersonix used high performance compute to run high-speed fluid dynamics simulations using Amazon EC2.

Overview | Opportunity | Solution | Outcome | AWS Services Used

1600

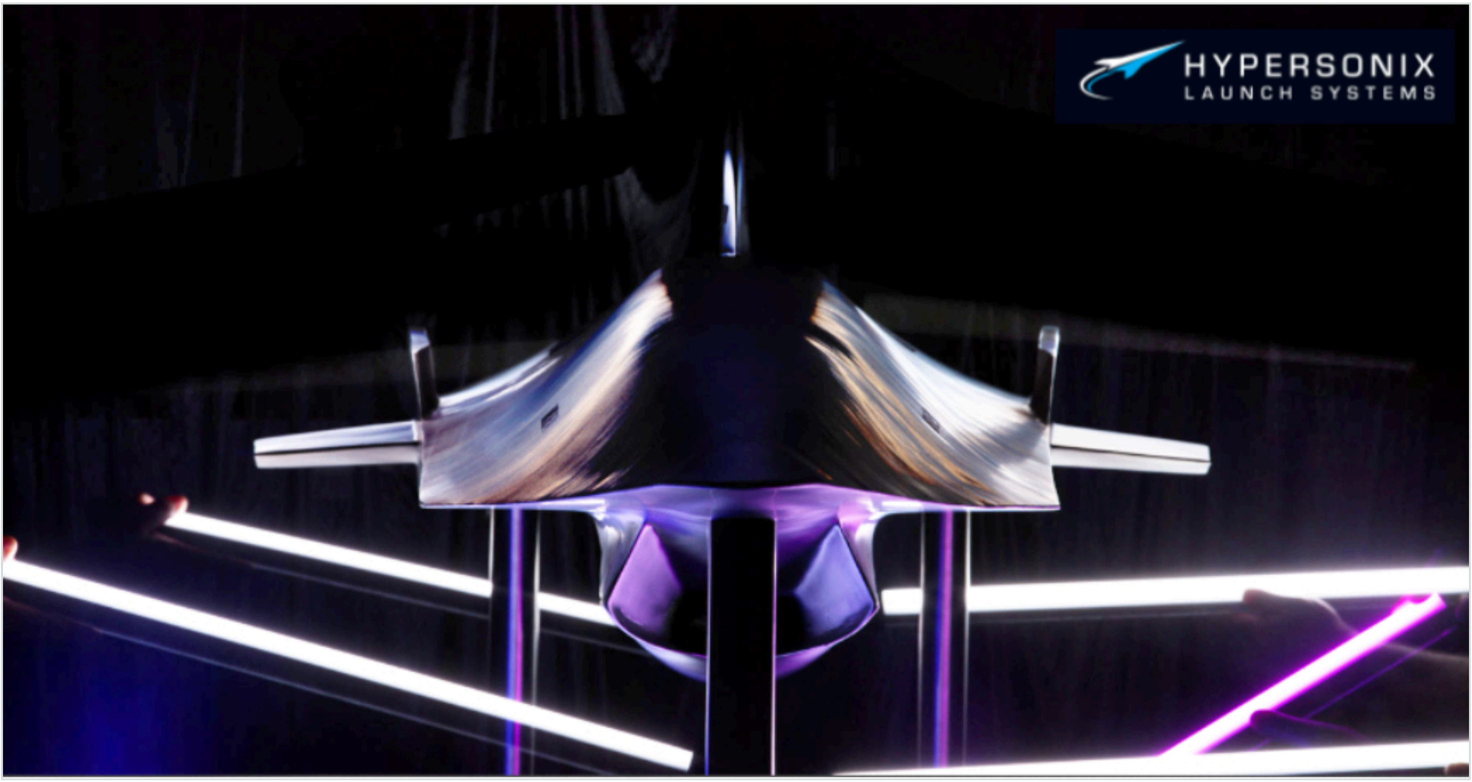
simulations run in 1 week

6-month

migration completed

Overview

Through the design of its hypersonic vehicle DART, Hypersonix Launch Systems (Hypersonix) needed to harness fast and scalable high performance compute (H
highly complex arena. The company turned
compute power it needed to improve its de
simulation runtime and reduced HPC costs.



Opportunity | Using Simcenter STAR-CCM+ on AWS to Scale HPC for Hypersonix

Founded in 2019, Hypersonix is an Australian company that aims to push the boundaries of sustainable hypersonic flight. To achieve this, the company is building a hydrogen-powered scramjet, a special type of engine that combusts fuel while the airflow through the engine moves faster than the speed of sound. Hypersonic aircraft will make it possible for airlines, freighters, and militaries to transport people and equipment at speeds above Mach 5 (greater than 1.5 km/sec). At this speed, continents can be crossed in tens of minutes.

What distinguishes Hypersonix's vehicles is that they are built with reusable parts and use hydrogen as fuel, which means that they don't create pollution during flight. "The aerospace industry creates a lot of pollution," says Dr. Michael Smart, cofounder, chief technology officer, and head of engineering at Hypersonix. "Being green is important to our company."

Building and testing these innovative aircraft requires HPC to create in-depth models and perform computational fluid dynamics (CFD) simulations. "Hypersonic flight is difficult and costly to experiment with," says Dr. Stephen Hall, head of advanced CFD thermal structural simulation at Hypersonix. "We rely on this computing and advanced simulation approach to aid our design and improve our vehicle." For its simulations, Hypersonix uses Simcenter STAR-CCM+, CFD software developed by Siemens Digital Industries Software, an AWS Partner. Thus, Hypersonix benefits from thousands of latest-generation CPUs, many terabytes of memory, and fast networking.

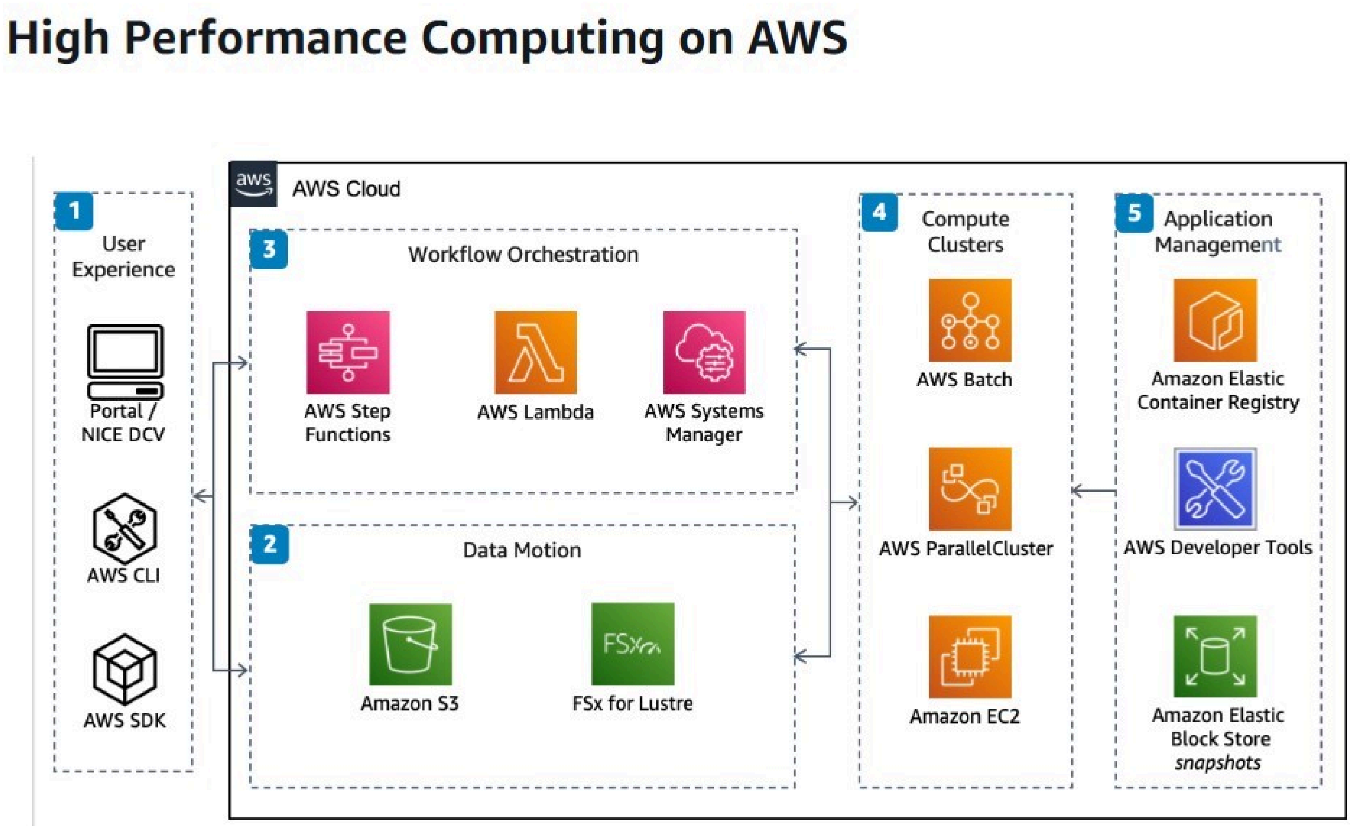
The HPC needs of Hypersonix—particularly for high fidelity and speed—quickly outgrew the company's on-premises 64-core workstation. The company knew that scaling on premises would be expensive and time-consuming. So by 2022, Hypersonix decided to migrate to AWS to securely and cost-effectively meet its needs. "Simcenter STAR-CCM+ has the flexibility to efficiently use all the different computing resources that AWS offers," says Dr. Hall. "The two make a very good combination and empower a world-class simulation capability."

Solution | Completing 1,600 Simulations in 1 Week Using Amazon EC2

Hypersonix migrated to AWS over 6 months, beginning in 2022. The Hypersonix team had never used AWS, but with close support from the AWS team, it soon overcame the learning curve to get its simulation processes set up and working efficiently.

To run its CFD workloads, Hypersonix used AWS Graviton3E-based Amazon Elastic Compute Cloud (Amazon EC2) Hpc7g Instances, which are powered by custom-designed AWS Graviton server processors developed by AWS. They use the most modern ARM chips and energy efficient green computing to empower excellent price performance for Star-CCM+ simulation HPC workloads on AWS. The company runs these instances through AWS ParallelCluster, an open-source cluster management tool that makes it simple to deploy and manage HPC clusters on AWS. After migration, the company saw its computational speed increase significantly. Hypersonix completed 1,600 simulations, each involving more than 70 million cells, in 1 week—a task that would have taken up to 3 months on its previously used on-premises workstations. “Part of innovation is solving issues quickly,” says Dr. Smart. “Using AWS, we manage to get answers and move fast with our simulations.”

Architecture Diagram



Click to enlarge for fullscreen viewing.

The company saved thousands of dollars during simulations by reducing simulation runtime and achieving greater efficiency and scalability. Simcenter STAR-CCM+ can scale very well, achieving cost-per-job parity for a wide range of core counts. Hypersonix can effectively double its cores on AWS, halving solution time, without increasing the cost per job. “We have already saved 3 months on the engineering design timeline for our first hypersonic aircraft for an AWS HPC investment of AUD 8,000 per month,” says Andy Mulholland, head of commercial and strategy at Hypersonix. “This is a huge benefit for us, especially because we’re on a startup budget and a very fast timeline to get to our first flight.” This empowered Hypersonix to rapidly provide designers with required simulation data and drive its design process forward. Hypersonix has not yet seen a drop-off in this ever-increasing rate of simulation and design iteration

In addition, the company uses Amazon FSx for Lustre, fully managed shared storage with the scalability and performance of the parallel disk Lustre file system. Thus, Hypersonix can reduce its save times from half an hour to a few minutes. “As we’re building the scramjet, time is money,” says Dr. Hall. “On AWS, we don’t have to factor in the time it takes to run the simulations anymore.”

Security was also an important factor in Hypersonix’s migration to AWS. In an industry that is not only competitive regarding intellectual property but also closely involved with militaries and export restrictions, Hypersonix needed to meet high standards for security and data control. AWS services facilitate compliancewith various security standards, including multi-factor authentication. Hypersonix also benefits from secure boundaries that limit access to its instances in specified zones under the AWS shared responsibility model.

To store more than 20 terabytes of computing data per month, the company secured its data sing Amazon Simple Storage Service (Amazon S3)—an object storage service offering industry-lea data availability, security, and performance. “Using AWS solutions gives us peace of mind that our work is secure.” says Dr. Hall.

Outcome | Using AWS to Pursue Speed and Innovation

The debut flight for Hypersonix’s vehicle is scheduled for early 2025. By migrating to AWS, Hypersonix gained the scalability to improve its simulations and pursue innovation. In the future, the company will enhance the quality and reliability of the data that is used in its models to run even more accurate and higher-resolution simulations.

Hypersonix’s next step is the HyperTwin X project, led by Mulholland, which encompasses a digital twin of its hypersonic vehicle using machine learning and artificial intelligence on AWS. The company also plans to migrate its onsite postprocessing and analysis to the cloud to reduce download time, improve collaboration, and minimize on-premises infrastructure.

As its computational and data-storage needs grow, Hypersonix is certain that it can push forward sustainably on AWS. “I believe that we can stand out from larger companies because we have the cloud capability and resources that we need on AWS,” says Dr. Hall.

About Hypersonix Launch Systems

Founded in 2019, Hypersonix Launch Systems builds and operates reusable and hydrogen-fueled hypersonic products capable of reaching Mach 7. Its sustainable flight technology is set to disrupt the global aerospace sector.

AWS Services Used

Amazon EC2 Hpc7g Instances

Amazon Elastic Compute Cloud (Amazon EC2) Hpc7g instances enable the best price performance for high performance computing (HPC) workloads on AWS. They offer up to 70% better performance and almost 3x better price performance compared to previous-generation AWS Graviton-based instances for compute-intensive HPC workloads.

Learn more »

AWS Parallel Cluster

AWS ParallelCluster is an open source cluster management tool that makes it easy for you manage High Performance Computing (HPC) clusters on AWS.

Learn more »

Amazon FSx for Lustre

Amazon FSx for Lustre provides fully managed shared storage with the scalability and performance of the popular Lustre file system.

Learn more »

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance. Millions of customers of all sizes and industries store, manage, analyze, and protect any amount of data for virtually any use case, such as data lakes, cloud-native applications, and mobile apps.

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